Introduction to Corporate Finance

会计学专业导论:

课程码 XCT5RW

课堂信息

- 2025级会计专业导论
- 2026年9月7日，13：20-16：30，文翠M-135
- 教师：刘宁悦，副教授，博士生导师，会计学（中外合作办学）项目责任教授
- 办公：文萃L402，81381103
- 邮箱：ningyueliu@bit.edu.cn

教师简历

Alliance Manchester Business School

Visiting Scholar

Michael Smurfit Business School, University College Dublin

Ph.D. in Finance

International Development Department, University of Oxford

Visiting Scholar

School of Business, Renmin University of China

Ph.D. in Financial Management

School of Business, Renmin University of China

B.A. in Business Administration

教学及参考用书

《财务管理学（第三版）》，刘宁悦、佟岩、肖淑芳，北京理工大学出版社，2023

《财务管理（第六版）》，王化成等，中国人民大学出版社，2020年
《公司理财》，（美）罗斯等，吴世农等译，机械工业出版社，2017年

Outline

- 什么是财务管理？
- 财务报表分析
- 资金的时间价值



课程框架

- **一个工具**
  - 财务报表分析
- **两个观念**
  - 时间价值
  - 风险报酬
- **四项内容**
  - 长期投资
  - 长期筹资
  - 分配政策
  - 营运资本管理

What is corporate finance?

- Corporate finance is to describe activities, decisions and techniques that deal with many aspects of a company's finances and capital.

- The Balance Sheet offers insight into the array of decisions, activities and objectives of the Financial Manager.

Balance Sheet Model of the Firm

Total Value of Assets:

Current Assets

Fixed Assets

1 Tangible

2 Intangible

Total Firm Value to Investors:

Current Liabilities

Long-Term Debt

Shareholders' Equity



The top three concerns of corporate finance

1. What long-term investments should the firm choose?
2. How should the firm raise funds for the selected investments?
3. How should current assets be managed and financed?

The Capital Budgeting Decision

Current Assets

Fixed Assets

1 Tangible

2 Intangible

What long-term investments should the firm choose?

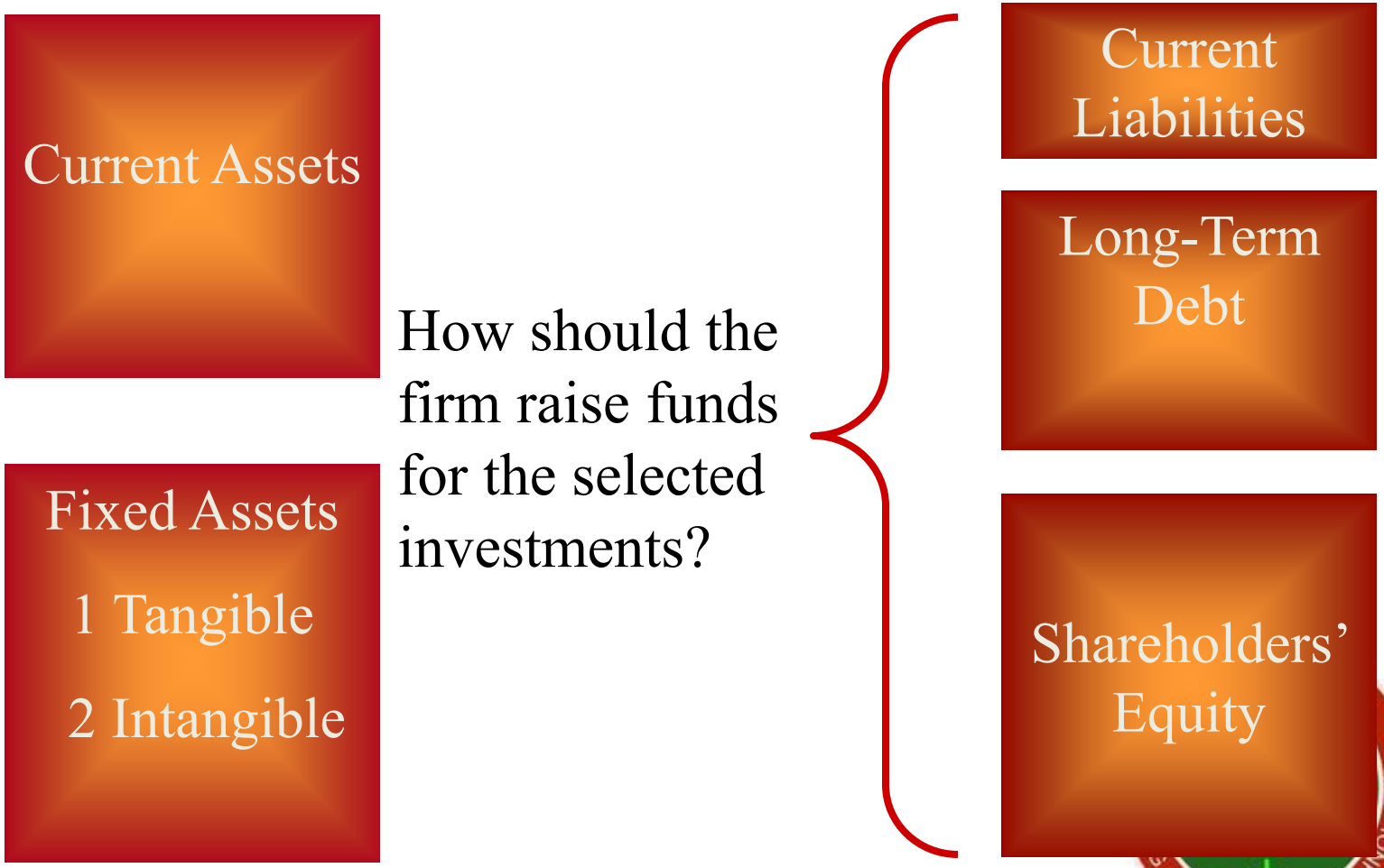
Current Liabilities

Long-Term Debt

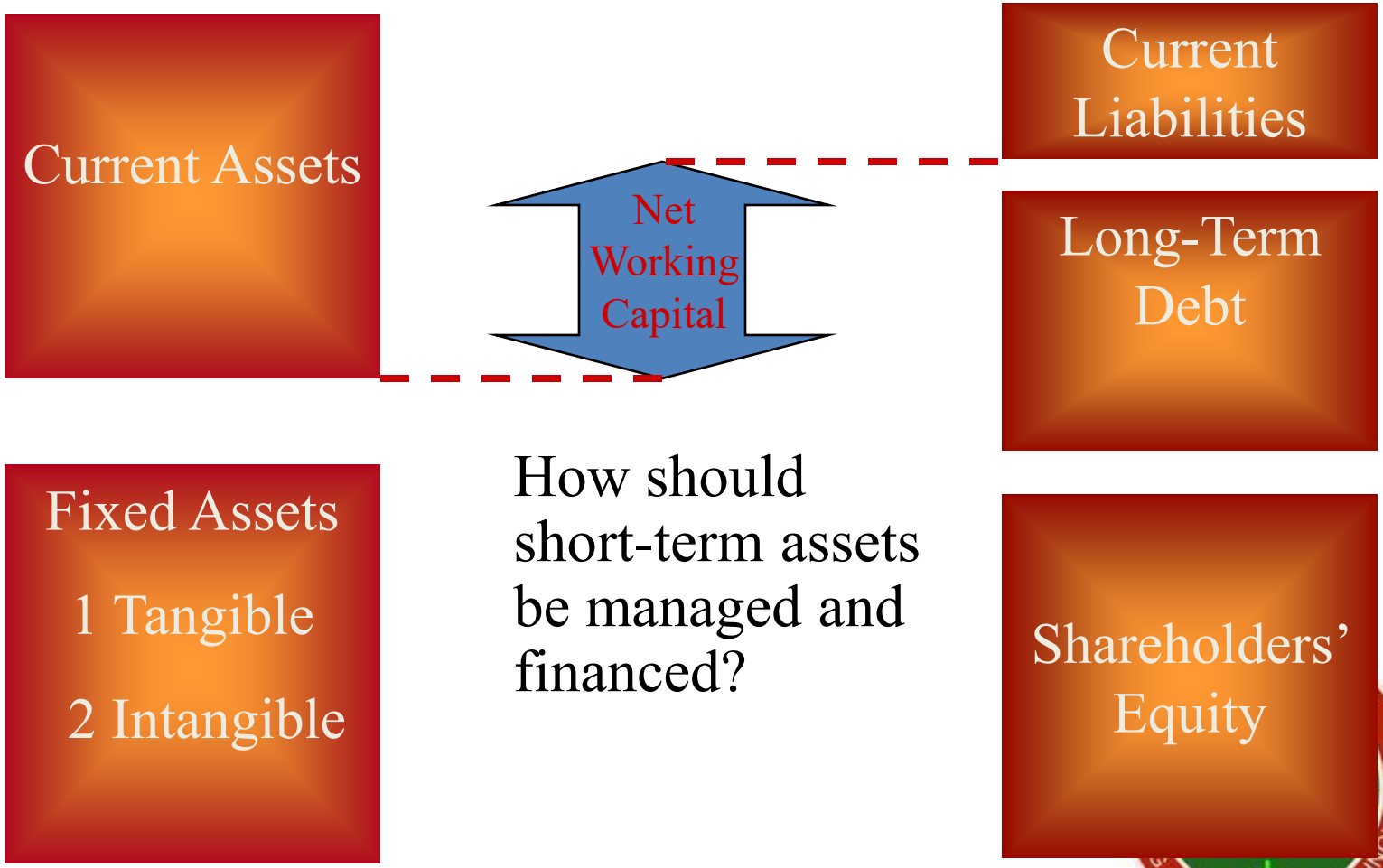
Shareholders' Equity



The Capital Structure Decision



Short-Term Asset Management



The corporate firm

- First company problem: raise funds
- The corporate form of business is the standard method for solving the problems encountered in raising large amounts of cash.
- There basic legal forms of organizing firms



Forms of Business Organization

- The Sole Proprietorship: owned by one person
- The Partnership
  - General Partnership: unlimited liability for all debts
  - Limited Partnership: at least one partner be a general partner and the limited partners do not participate in managing the business (to be limited to the amount of cash each has contributed to the partnership)
- Advantage: the cost of getting started
- Disadvantage: unlimited liability, limited life of the enterprise, difficulty of transferring ownership, difficulty in raising cash



The Corporation

- A corporation is a type of business organization that is recognized under the law as an entity separate from its owners. Therefore, the owners are not personally liable for the debts of the business. These owners can lose no more than the amounts they have invested in the business- a concept know as limited liability.

Corporation VS. Partnership

	Corporation	Partnership
Liquidity and marketability	Shares can be easily exchanged	Subject to substantial restrictions
Voting Rights	Usually each share gets one vote	General Partner is in charge; limited partners may have some voting rights
Taxation	Double	Partners pay taxes on distributions
Reinvestment and dividend payout	Broad latitude	All net cash flow is distributed to partners
Liability	Limited liability	General partners may have unlimited liability; limited partners enjoy limited liability
Continuity	Perpetual life	Limited life

The goal of financial management

- What is the correct goal?
  - Maximize output?
  - Maximize income?
  - Maximize market share?
  - **Maximize shareholder wealth?**
  - **The goal of financial management is to maximize the current value per share of the existing stock.**



The Agency Problem

- Agency relationship
  - Principal hires an agent to represent his/her interest
  - Stockholders (principals) hire managers (agents) to run the company
- Agency problem
  - Conflict of interest between principal and agent

Agency Cost

- Cost of Conflict of Interest
- Example:
  - Large investment positions firm for long term positive cash flow but has risk in short run
    - Owners want this investment – Increases firm value
    - Managers object – Risk may have personal cost
  - If managers prevail, foregone long term cash flow is the Agency Cost

Managerial Goals

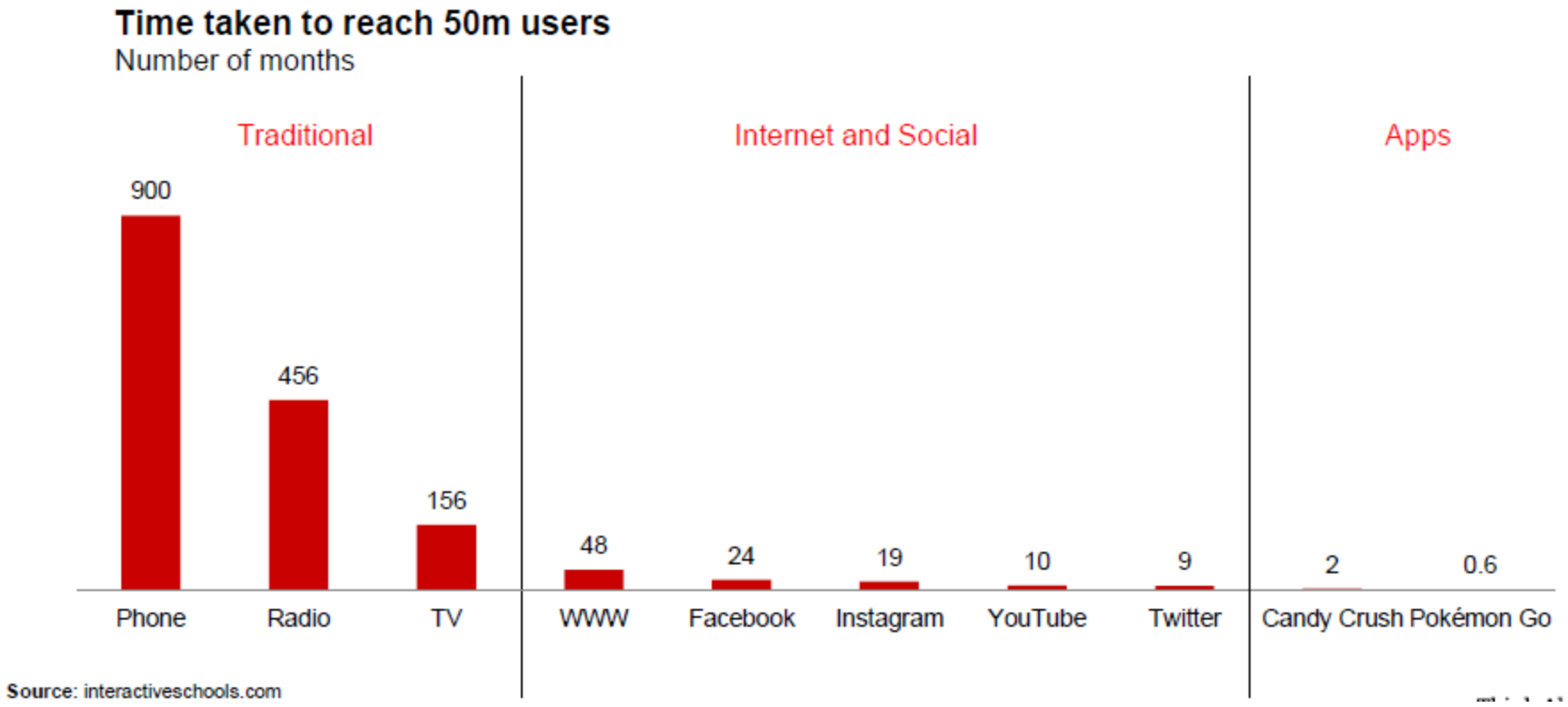
- Managerial goals may be different from shareholder goals
  - Expensive perquisites
  - Survival
  - Independence
- Increased growth and size
  - Often lead to management reward
  - Not necessarily in best interest of shareholders

Managing Managers

- Managerial compensation
  - Incentives can be used to align management and stockholder interests
  - The incentives need to be structured carefully to make sure that they achieve their intended goal
- Corporate control
  - The threat of a takeover may result in better management
- Influence of other stakeholders

Financial Management Environment

- The accelerating pace of change



ABCD

- Artificial Intelligence
- Blockchain
- Cloud
- Big Data

- Robotic Process Automation
- https://www.uipath.com/

Financial Statements Analysis



Outline

- Financial Statements Analysis
- Ratio Analysis
- The DuPont Identity
- Using Financial Statement Information



Learning Objectives

- LO1 How to standardize financial statements for comparison purposes;
- LO2 How to compute and, more importantly, interpret some common ratios;
- LO3 The determinants of a firm's profitability;
- LO4 Some of the problems and pitfalls in financial statement analysis.



Financial Statements Analysis

- Making financial statements useful for users is one finance role
- In order to make meaningful comparisons of companies of different size, financial professionals use two key techniques:
  - Common-Size Statements
  - Financial Ratios

Common-Size Financial Statements

- Common-Size Balance Sheets
  - Compute all accounts as a percent of total assets
- Common-Size Income Statements
  - Compute all line items as a percent of sales
- Standardized statements make it easier to compare financial information, particularly as the company grows.
- They are also useful for comparing companies of different sizes, particularly within the same industry.



Prufrock Corporation
Comparison: Financial & Common Size Balance Sheet

PRUFROCK CORPORATION
Balance Sheets as of December 31, 2009 and 2010
($ in millions)

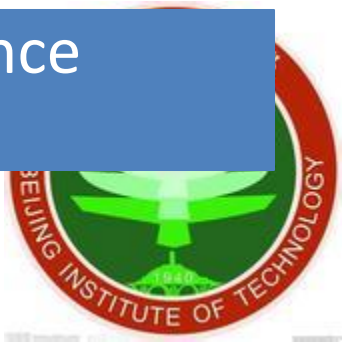
Assets	2009	2010
Current assets		
Cash	$ 84	$ 98
Accounts receivable	165	188
Inventory	393	422
Total	$ 642	$ 708
Fixed assets		
Net plant and equipment	$ 2,731	$ 2,880
Total assets	$ 3,373	$ 3,588
Liabilities and Owners' Equity		
Current liabilities		
Accounts payable	$ 312	$ 344
Notes payable	231	196
Total	$ 543	$ 540
Long-term debt	$ 531	$ 457
Owners' equity		
Common stock and paid-in surplus	$ 500	$ 550
Retained earnings	1,799	2,041
Total	$ 2,299	$ 2,591
Total liabilities and owners' equity	$ 3,373	$ 3,588

PRUFROCK CORPORATION
Common-Size Balance Sheets
December 31, 2009 and 2010

Assets	2009	2010	Change
Current assets			
Cash	2.5%	2.7%	+ .2%
Accounts receivable	4.9	5.2	+ .3
Inventory	11.7	11.8	+ .1
Total	19.0	19.7	+ .7
Fixed assets			
Net plant and equipment	81.0	80.3	− .7
Total assets	100.0%	100.0%	.0%
Liabilities and Owners' Equity			
Current liabilities			
Accounts payable	9.2%	9.6%	+ .4%
Notes payable	6.8	5.5	−1.3
Total	16.1	15.1	−1.0
Long-term debt	15.7	12.7	−3.0
Owners' equity			
Common stock and paid-in surplus	14.8	15.3	+ .5
Retained earnings	53.3	56.9	+3.5
Total	68.2	72.2	+4.1
Total liabilities and owners' equity	100.0%	100.0%	.0%

Balance Sheet

Common Size Balance Sheet



Prufrock Corporation
Comparison: Financial & Common Size Income Statement

Income Statement

PRUFROCK CORPORATION 2010 Income Statement ($ in millions)		
Sales		$2,311
Cost of goods sold		1,344
Depreciation		276
Earnings before interest and taxes		$ 691
Interest paid		141
Taxable income		$ 550
Taxes (34%)		187
Net income		$ 363
Dividends	$121	
Addition to retained earnings	242	

Common Size Income Statement

PRUFROCK CORPORATION Common-Size Income Statement 2010		
Sales		100.0%
Cost of goods sold		58.2
Depreciation		11.9
Earnings before interest and taxes		29.9
Interest paid		6.1
Taxable income		23.8
Taxes (34%)		8.1
Net income		15.7%
Dividends	5.2%	
Addition to retained earnings	10.5	

Ratio Analysis

- Ratios compliment common size analysis and allow for deeper comparison through time or between dissimilar companies
- Not always computed precisely the same; document your approach
- As we look at each ratio, ask yourself:
  - How is the ratio computed?
  - What is the ratio trying to measure and why?
  - What is the unit of measurement?
  - What does the value indicate?
  - How can we improve the company's ratio?



Categories of Financial Ratios

- Short-term solvency or liquidity ratios
- Long-term solvency or financial leverage ratios (Leverage and Coverage ratios)
- Asset management or turnover ratios
- Profitability ratios
- Market value ratios

- Example: Prufrock Corporation

Prufrock Corporation Balance Sheet

PRUFROCK CORPORATION Balance Sheets as of December 31, 2009 and 2010 ($ in millions)		
Assets	**2009**	**2010**
Current assets		
Cash	$ 84	$ 98
Accounts receivable	165	188
Inventory	393	422
Total	$ 642	$ 708
Fixed assets		
Net plant and equipment	$ 2,731	$ 2,880
Total assets	$ 3,373	$ 3,588
Liabilities and Owners' Equity		
Current liabilities		
Accounts payable	$ 312	$ 344
Notes payable	231	196
Total	$ 543	$ 540
Long-term debt	$ 531	$ 457
Owners' equity		
Common stock and paid-in surplus	$ 500	$ 550
Retained earnings	1,799	2,041
Total	$ 2,299	$ 2,591
Total liabilities and owners' equity	$ 3,373	$ 3,588



Prufrock Corporation Income Statement

PRUFROCK CORPORATION 2010 Income Statement ($ in millions)		
Sales		$2,311
Cost of goods sold		1,344
Depreciation		276
Earnings before interest and taxes		$ 691
Interest paid		141
Taxable income		$ 550
Taxes (34%)		187
Net income		$ 363
Dividends	$121	
Addition to retained earnings	242	



Liquidity Ratios

- Current Ratio = CA / CL
  - 708 / 540 = 1.31 times
- Quick Ratio = (CA – Inventory) / CL
  - (708 - 422) / 540 = .53 times
- Cash Ratio = Cash / CL
  - 98 / 540 = .18 times

Other Liquidity Ratios

- Net working capital to total assets= NWC / TA
  - (708-540) / 3,588 = 4.7%
- Interval measure= CA / Average daily operating costs
  - 708 / (1,344/365) = 708/ 3.68 = 192 days
  - Prufrock could hang on for six months or so!



Leverage Ratios

- Total Debt Ratio = (TA – TE) / TA
  - (3588 - 2591) / 3588 = .28 times
- Debt-Equity Ratio = TD / TE
  - (3588 – 2591) / 2591 = .38 times
- Equity Multiplier = TA / TE = 1 + D/E
  - 1 + .38 = 1.38 times
- Long-term Debt Ratio= Long-term debt / (Long-term debt + TE)
  - 457 / (457+2,591) = .15 times



Coverage Ratios

- Times Interest Earned Ratio = EBIT / Interest
  - 691 / 141 = 4.9 times
- Cash Coverage = (EBIT + Depreciation + Amortization) / Interest
  - (691 + 276) / 141 = 6.9 times



Inventory Ratios

- Inventory Turnover = Cost of Goods Sold / Inventory
  - 1,344 / 422 = 3.2 times
- Days' Sales in Inventory = 365 / Inventory Turnover
  - 365 / 3.2 = 115 days
- Ending figure VS. Average figure
- Inventory Turnover = Cost of Goods Sold / Average Inventory
  - 1,344 /[(393+422) /2]= 3.3 times

Receivables Ratios

- Receivables Turnover = Sales / Accounts Receivable
  - 2,311 / 188 = 12.3 times
- Days' Sales in Receivables = 365 / Receivables Turnover
  - 365 / 12.3 = 30 days
- This ratio is frequently called the average collection period (ACP).



Payables Ratios

- Payables Turnover = Cost of Goods Sold / Accounts Payable
  - 1,344 / 344 = 3.9 times
- Days' Sales in Payables = 365 / Payables Turnover
  - 365 / 3.9 = 94 days
- Prufrock takes 94 days to pay while 30 days to receive.

Asset Turnover Ratios

- NWC Turnover = Sales / NWC
  - 2,311 / (708-540) = 13.8 times
- Fixed Asset Turnover = Sales / Net Fixed Assets
  - 2,311 / 2,880 = .80 times
- Total Asset Turnover = Sales / Total Assets
  - 2,311 / 3,588 = .64 times
  - It is not unusual for TAT < 1, especially if a firm has a large amount of fixed assets.



Account Receivable Management of JIT

Shenzhen JIT Technology Co., Ltd
集电科技（深圳）有限公司

Shenzhen JIT Technology Co., Ltd was established in 1999, we are providing a complete set of services to all electronic & electrical manufacturing suppliers. The Company's mission is to provide a series of chemical materials, industry precision sticker’s products and all kinds of specifications equipment for both electronic & electrical manufacturing suppliers.

Company's core product: Insulation Material; Coating & Dispensing Machine; Adhesive; Silicone Material; Potting Material; Tape & Adhesive; Lead free soldering tool; Liquid optical adhesive etc…

表 1 深圳集电科技有限责任公司资产负债表（2011 年 12 月）

BALANCE SHEET (From 1Dec 2011 to 31 Dec 2011)

UNIT:RMB

ASSETS	SUM	LIABILITIES AND CAPITAL OF SHAREHOLDERS	SUM
CURRENT ASSETS:		CURRENT LIABILITIES:	
CURRENCY:	1,524,199	ACCOUNTS PAYABLE	10,650,377
Cash	215,824	NOTES PAYABLE	0
Cash in Bank	1,308,375	ACCRUED EXPENSES	1,425,945
ACCOUNTS RECEIVABLE	27,977,006	Including: Commission	445,373
NOTES RECEIVABLE	328,864	Stomers Discount	100,219
OTHER ACCOUNTS RECEVIABLE	1,900,708	Bonus	516,918
RESERVE FOR BAD DEBTS	(1,011,271)	Mould	0
		Children Fund	196,472
INVENTORIES	9,906,814	PVA	37,804
		Exhibition Fees	129,159
Including: 3M	3,380,639	Special Fees	0
JIT	2,013,211	OTHER ACCOUNTS PAYABLE	1,683,834
BECTRON	1,896,625	TAXATION PAYABLE	436,351
Lord	62,167	SHORTERM LOAN	
Diffuser	1,425,836	LONGTERM LOAN	2,364,281
KCC	884	TOTAL CURRENT IABILITIES	16,560,789
Hunstman	117,761	CAPITAL OF SHAREHOLDERS:	
CRC	21,420	PAID-UP CAPITAL	3,000,000
PVA	988,270		
LONG-TERM INVESTMENT	944,247	Including: Qualitek (50%) Paid	1,500,000
LONG-PREPAID EXPENSES	1,039,234	Bosny (43%) Paid	1,290,000
PREPAID EXPENSES(Factory)	93,529	Bernard (7%)	210,000
EXCESS INVENTORY	1,578,892	NET VALUE OF CAPITAL	3,000,000
TOTAL CURRENT ASSETS	44,282,223	NET PROFIT ON 2011	4,748,265
FIXED ASSETS:		Poor Children Fund (Profit 1%)	(47,000)
VALUE OF FIXED ASSETS	520,847	NET PROFIT ON 2010	4,654,176
VALUE OF FIXED ASSETS	3,129,165	Poor Children Fund (Profit 1%)	(46,000)
TOTAL FIXED ASSETS	3,650,012	NET PROFIT ON 2009	3,052,961
LESS: DEPRECIATION(Factory)	1,267,082	Poor Children Fund(Profit 1%)	(32,960)
LESS: DEPRECIATION	289,150	UNDISTRIBUTED PROFIT(2003-2008)	15,187,524
TOTAL DEPRECIATION	1,556,232	DISTRIBUTED PROFIT	(701,752)
NET VALUE OF FIXED ASSETS	2,093,780	TOTAL PROFIT	26,815,214
TOTAL FIXED ASSETS	2,093,780	TOTAL CAPITAL OF SHAREHOLDERS	29,815,214
TOTAL ASSETS	46,376,003	TOTAL LIABILITIES AND CAPITAL OF SHAREHOLDERS	46,376,003

表 2 深圳集电科技有限责任公司资产负债表（2010 年 12 月）

BALANCE SHEET (From 1 Dec 2010 to 31 Dec 2010)

UNIT:RMB

ASSETS	SUM	LIABILITIES AND CAPITAL OF SHAREHOLDERS	SUM
CURRENT ASSETS:		CURRENT LIABILITIES:	
CURRENCY:	2,169,940	ACCOUNTS PAYABLE	10,916,254
Cash	603,134	NOTES PAYABLE	66,400
Cash in Bank	1,566,807	ACCRUED EXPENSES	1,150,194
ACCOUNTS RECEIVABLE	20,637,946	Including: Commission	357,903
NOTES RECEIVABLE	564,979	Exhibition Fees	4,461
OTHER ACCOUNTS RECEVIABLE	1,180,280	Customers Discount	145,926
RESERVE FOR BAD DEBTS	0	Bonus	448,377
		Special Fees	13,444
INVENTORIES	14,424,976	Children Fund	154,472
		PVA	19,201
Including: 3M	6,996,820	Mould	6,410
Weller	1,582,553	OTHER ACCOUNTS PAYABLE	2,524,124

BECTRON	1,723,053	TAXATION PAYABLE	(321,615)
Lord	64,411	SHORTERM LOAN	
KCC	884	LONGTERM LOAN	1,689,940
Diffuser	3,649,746	TOTAL CURRENT LIABILITIES	16,025,298
Hunstman	103,181	CAPITAL OF SHAREHOLDERS:	
CRC	35,790	PAID-UP CAPITAL	3,000,000
PVA	268,538		
LONG-TERM INVESTMENT	846,182	Including: Qualitek 60% Paid	1,500,000
PREPAID EXPENSES(Factory)	0	Bosny (43%) Paid	1,290,000
EXCESS INVENTORY	(780,000)	Bernard (7%)	210,000
TOTAL CURRENT ASSETS	39,044,303	NET VALUE OF CAPITAL	3,000,000
FIXED ASSETS:		NET PROFIT ON 2010	4,654,176
VALUE OF FIXED ASSETS	440,048	Poor Children Fund (Profit 1%)	(46,000)
VALUE OF FIXED ASSETS(Factory)	2,880,127	NET PROFIT ON 2009	3,052,961
TOTAL FIXED ASSETS	3,320,175	Poor Children Fund (Profit 1%)	(32,960)
LESS: DEPRECIATION(Factory)	970,582	NET PROFIT ON 2008	3,896,968
LESS: DEPRECIATION	254,650	Poor Children Fund (Profit 1%)	(37,662)
TOTAL DEPRECIATION	1,225,232	UNDISTRIBUTED PROFIT(2003-2007)	11,290,556
NET VALUE OF FIXED ASSETS	2,094,943	DISTRIBUTED PROFIT	(664,090)
		TOTAL PROFIT	22,113,949
TOTAL FIXED ASSETS	2,094,943	TOTAL CAPITAL OF SHAREHOLDERS	25,113,949
TOTAL ASSETS	41,139,246	TOTAL LIABILITIES AND CAPITAL OF SHAREHOLDERS	41,139,247

Table 2 Continue

中国管理案例共享中心

北京理工大学 BEIJING INSTITUTE OF TECHNOLOGY 1940

Accounts Receivable / Current Assets

- Accounts Receivable / Current Assets
  - 2011: 27,977,006 / 44,282,223 = 63.18%
  - 2010: 20,637,946 / 39,044,303 = 52.86%

Consequences of high level of accounts receivable

- High operating risk
- Lack of working capital
- Influencing enterprise credit “Chain Debts”

- **Improvements**
- For clients:

➢Normal credit terms:“1/10、N/30”, “2/10、1/20、N/30” or“3/30、2/45、N/60”

➢A way of providing cash discounts on purchases. It means that if the bill is paid within 10 days, there is a 1% discount. Otherwise, the total amount is due within 30 days.

➢JIT: “2/60、1/90、N/180”

- **Improvements**
- For employees:

- Reward fomula = clearing contract value × saving days × interest rate of bank loan/2/365

- Days' Sales in Receivables in 2010 is 280 and in 2011 is 257. The average Days' Sales in Receivables is 268.5. JIT sets 270 days as standard.

- For employee A, he made a deal with company X on 25th April. The contract value is 530000 Yuan. Company X paid 100000 on 25th April. The credit term is "2/60、1/90、 n/180". Company X paid 425700 on 24th July (430000 × 1% =4300, 430000-4300=425700). The interest rate of bank loan is 6.5%.

- Reward of A = 425700 × (270-90) × 6.5% × 0.5 × 1/365=6822.86 Yuan

- **Improvements**
- Accounts Receivable Financing

- Accounts receivable financing is a type of asset-financing arrangement in which a company uses its receivables - which is money owed by customers - as collateral in a financing agreement.
- The company receives an amount that is equal to a reduced value of the receivables pledged. The age of the receivables have a large effect on the amount a company will receive. The older the receivables, the less the company can expect.

Profitability Measures

- Profit Margin = Net Income / Sales
  - 363 / 2,311 = 15.7%
- EBITDA Margin = EBITDA / Sales
  - 967 / 2,311 = 41.8%
- Return on Assets (ROA) = Net Income / Total Assets
  - 363 / 3,588 = 10.12%
  - Return on book assets
- Return on Equity (ROE) = Net Income / Total Equity
  - 363 / 2,591 = 14.0%
  - Return on book equity or return on net worth



Market Value Measures

- We assume that Prufrock has 33 million shares outstanding and the stock sold for $88 per share at the end of the year.
- Earnings per share = Net income / Shares outstanding
  - 363 / 33 = $11
- PE Ratio = Price per share / Earnings per share
  - 88 / 11 = 8 times
  - PE ratio measures how much investors are willing to pay per dollar of current earnings.

Market Value Measures

- Suppose Prufrock's anticipated growth rate in EPS was 6 percent.
- PEG ratio = PE ratio / (expected future earnings growth rates * 100)
  - 8 / 6 = 1.33
  - High PEG ratios suggest that the PE is too high relative to growth, and vice versa.
- Price-sales Ratio = Price per share / Sales per share
  - 88 / (2,311/33) = 88 / 70 = 1.26
  - A recent start-up may have negative earnings for extended periods.



Market Value Measures

- Market-to-book ratio = market value per share / book value per share
  - 88 / (2,591 / 33) = 1.12 times
  - A value less than 1 could mean that the firm has not been successful overall in creating value for its stockholders.
- Tobin's Q = market value of firm's assets / Replacement cost of firm's assets
  - = market value of firm's debt and equity / Replacement cost of firm's assets
  - It is accuracy but difficult to estimate the replacement cost of a firm's assets. Book values can be used instead in some cases, but accuracy may suffer.

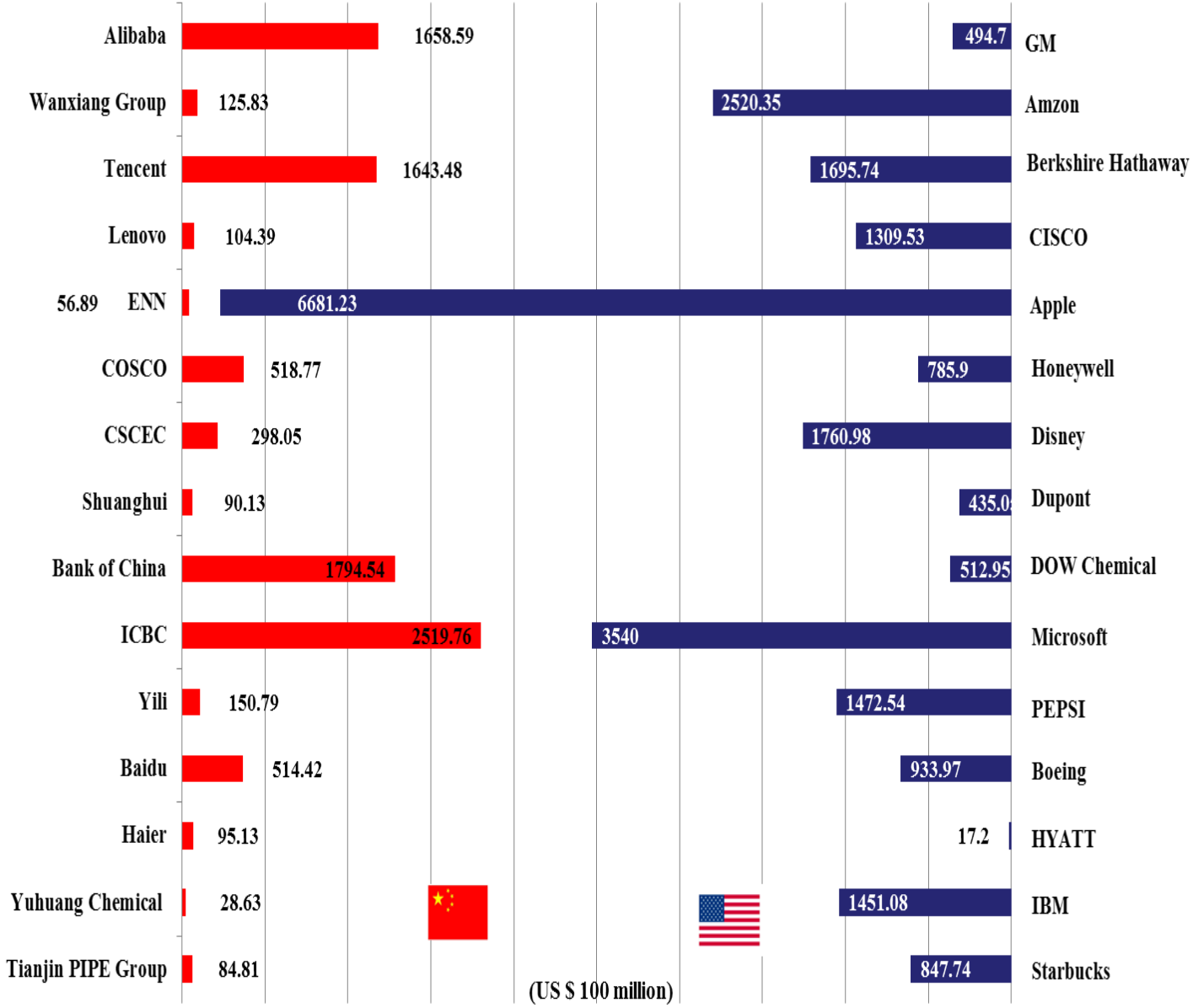
Market Value Measures

- Market Capitalization = Price per share x Shares outstanding
  - $88 per share x 33 million shares = 2,904 million
- Enterprise Value (EV) = Total market value of the stock + Book value of all liabilities – Cash
  - EV is an estimate of the market value of the company's operating assets. Operating assets are all the assets of the firm except cash.
  - We can use the right-hand side of the balance sheet to instead individual assets.
  - We use the book value for liabilities because we typically can not get the market values.
- EV Multiple = EV / EBITDA



President Xi Jinping made his state visit to America from Sep.22 to Sep.28. He met 30 CEOs or Chairmen of Chinese and America company which were participating in US-China Entrepreneur Roundtable Dialogue during his visit to Microsoft on Sep. 23,2015.

Market Capitalization: China-US Entrepreneurs Roundtable



Datas from Dec. 2014

15 Chinese Enterprises in US



INTERNET

Alibaba: It was chosen to listed on New York Stock Exchange and to be the biggest IPO in U.S. history in 2014. Now, its first overseas cloud-computing data center has been working in Silicon Valley since March, 2015.

Tencent: Developing new business actively in North America, like Wechat.

Baidu: Go public on Nasdaq in 2005, the world's largest Chinese-language search engine .



TECHNOLOGY

Lenovo: Purchased IBM's PC business and agreed with Google to acquire Motorola.



FINANCE

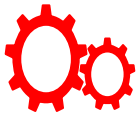
ICBC: Take over the Prime Dealer Services of Fortis Securities and BEA(U.S.A.)

Bank of China: Offer $250 million real estate loans in North America in 2014



AUTOMOBILES

Wanxiang: Involved in automotive and industrial markets in the U.S.



MANUFACTURING

Haier: It's in the first camp of Appliance Industry in North America Market

Tianjin PIPE: Set a subsidiary in Huston, Texas with supporting seamless steel pipe.



FOOD INDUSTRY

Shuanghui: Purchased 100 percent stake in Smithfield Foods Inc, the largest U.S. pork and hog producer.

Yili: Begin to set up overseas subsidiaries.



CONSTRUCTION

CSCEC: Kick-start high-speed-rail projects in western California with CRRC (China Railway Rolling Stock Corporation)



LOGISTIC INDUSTRY

COSCO: Setting up offices across North America

伯克希尔哈撒韦持仓数据（截至2020年12月31日）

公司名称	持股数量	占公司持股百分比(%)	购买价格（百万美元）	市场价值（百万美元）
苹果	907,559,761	5.4	31,089	120,424
美国银行	1,032,852,006	11.9	14,631	31,306
可口可乐	400,000,000	9.3	1,299	21,936
美国运通	151,610,700	18.8	1,287	18,331
Verizon通信	146,716,496	3.5	8,691	8,620
穆迪	24,669,778	13.2	248	7,160
美国合众银行	148,176,166	9.8	5,638	6,904
比亚迪	225,000,000	8.2	232	5,897
雪佛龙	48,498,965	2.5	4,024	4,096
Charter通信	5,213,461	2.7	904	3,449
纽约梅隆银行	66,835,615	7.5	2,918	2,837
艾伯维公司	25,533,082	1.4	2,333	2,736
默克	28,697,435	1.1	2,390	2,347
伊藤忠商事	81,304,200	5.1	1,862	2,336
通用集团	52,975,000	3.7	1,616	2,206

大数投资

Summary of Ratio Formulae

I. Short-Term Solvency, or Liquidity, Ratios

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}}$$

$$\text{Cash ratio} = \frac{\text{Cash}}{\text{Current liabilities}}$$

II. Long-Term Solvency, or Financial Leverage, Ratios

$$\text{Total debt ratio} = \frac{\text{Total assets} - \text{Total equity}}{\text{Total assets}}$$

$$\text{Debt-equity ratio} = \text{Total debt/Total equity}$$

$$\text{Equity multiplier} = \text{Total assets/Total equity}$$

$$\text{Times interest earned ratio} = \frac{\text{EBIT}}{\text{Interest}}$$

$$\text{Cash coverage ratio} = \frac{\text{EBITDA}}{\text{Interest}}$$

III. Asset Utilization, or Turnover, Ratios

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Inventory}}$$

$$\text{Days' sales in inventory} = \frac{\text{365 days}}{\text{Inventory turnover}}$$

$$\text{Receivables turnover} = \frac{\text{Sales}}{\text{Accounts receivable}}$$

$$\text{Days' sales in receivables} = \frac{\text{365 days}}{\text{Receivables turnover}}$$

$$\text{Total asset turnover} = \frac{\text{Sales}}{\text{Total assets}}$$

$$\text{Capital intensity} = \frac{\text{Total assets}}{\text{Sales}}$$

IV. Profitability Ratios

$$\text{Profit margin} = \frac{\text{Net income}}{\text{Sales}}$$

$$\text{Return on assets (ROA)} = \frac{\text{Net income}}{\text{Total assets}}$$

$$\text{Return on equity (ROE)} = \frac{\text{Net income}}{\text{Total equity}}$$

$$\text{ROE} = \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}$$

V. Market Value Ratios

$$\text{Price-earnings ratio} = \frac{\text{Price per share}}{\text{Earnings per share}}$$

$$\text{Market-to-book ratio} = \frac{\text{Market value per share}}{\text{Book value per share}}$$

$$\text{EV multiple} = \frac{\text{Enterprise value}}{\text{EBITDA}}$$



The DuPont Identity

- Popularized by the DuPont Corporation
- A more sophisticated method of evaluating return
- Illustrates the interaction between profit, assets and leverage
- Holds that ROE is actually a function of 3 measures:
  - Operating Efficiency (Profit Margin)
  - Asset Use Efficiency (Total Asset Turnover)
  - Financial Leverage (Equity Multiplier)



Derivation of The DuPont Identity

- ROE = NI / TE
- Multiply by 1 and then rearrange:
  - ROE = (NI / TE) (TA / TA)
  - ROE = (NI / TA) (TA / TE) = ROA * EM
- Multiply by 1 again and then rearrange:
  - ROE = (NI / TA) (TA / TE) (Sales / Sales)
  - ROE = (NI / Sales) (Sales / TA) (TA / TE)
  - ROE = PM * TAT * EM

Using The DuPont Identity

- ROE = PM * TAT * EM
  - Profit margin is a measure of the firm's operating efficiency – how well it controls costs.
  - Total asset turnover is a measure of the firm's asset use efficiency – how well it manages its assets.
  - Equity multiplier is a measure of the firm's financial leverage.



Calculating The DuPont Identity

- ROA = 10.12% and EM = 1.385
  - ROE = 10.12% * 1.385 = 14.0%
- PM = 15.7% and TAT = 0.64
  - ROE = 15.7% * 0.64 * 1.385 = 14.0%

The Du Pont Breakdown for General Motors

GM							
	ROE	=	PM	*	TAT	*	EM
1989	12.1%	=	3.4%	*	0.71	*	4.95
1993	44.1%	=	1.8%	*	0.73	*	33.62

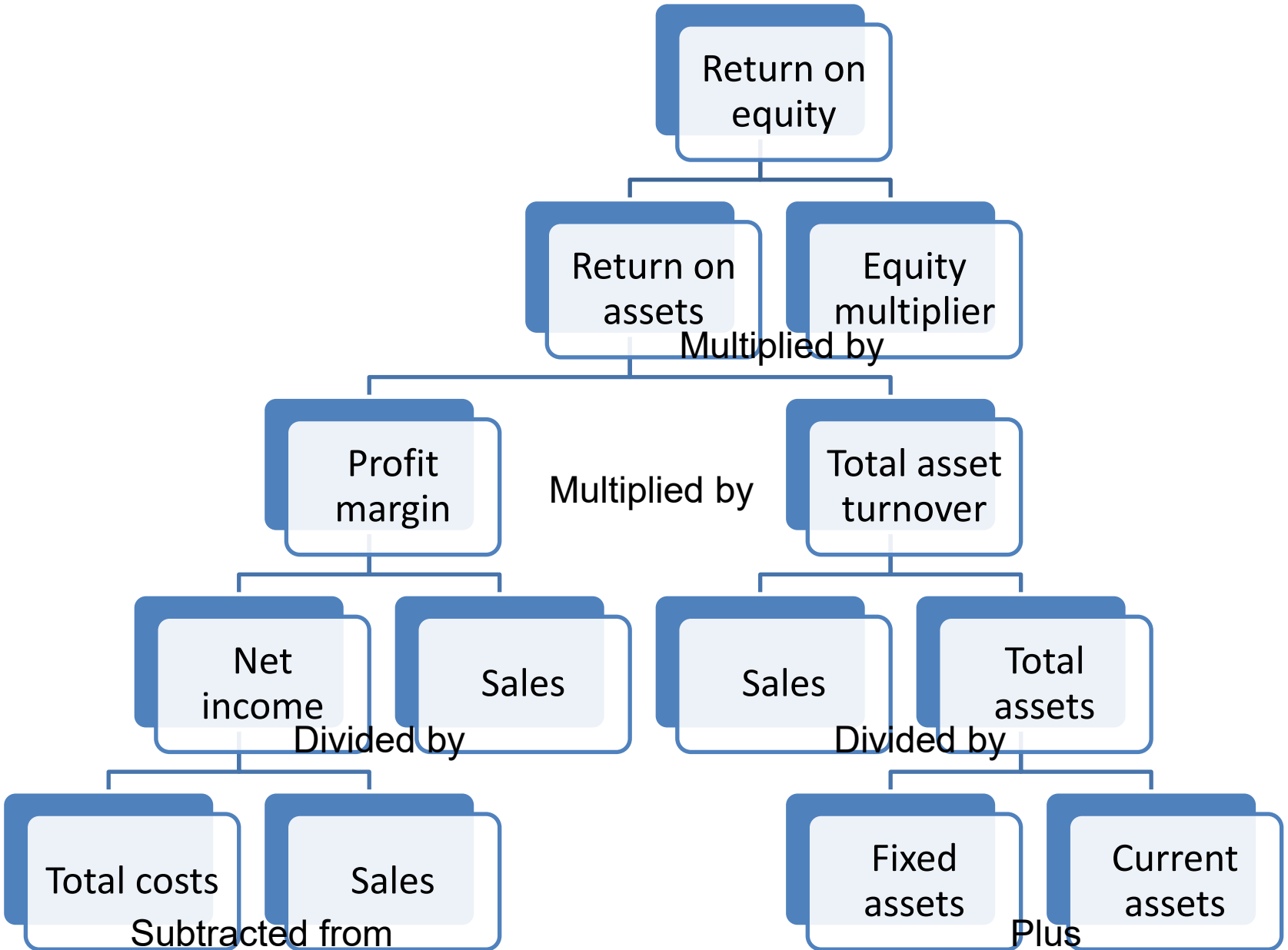
- How is it possible for GM's ROE to have climbed so sharply?
- In fact, what happened was that GM's book equity value was almost wiped out overnight in 1992 by changes in the accounting treatment of pension liabilities.
- Accounting change VS. financial performance

The Du Pont Breakdown for Yahoo! And Google

Yahoo!							
	ROE	=	PM	*	TAT	*	EM
12/09	4.8%	=	9.3%	*	0.433	*	1.20
12/08	3.8%	=	5.9%	*	0.527	*	1.22
12/07	6.9%	=	9.5%	*	0.570	*	1.28
Google							
	ROE	=	PM	*	TAT	*	EM
12/09	18.1%	=	27.6%	*	0.584	*	1.12
12/08	14.9%	=	19.4%	*	0.686	*	1.12
12/07	18.6%	=	25.3%	*	0.655	*	1.12



Extended DuPont Chart



Using Financial Ratios

- Ratios are not very helpful by themselves: they need to be compared to something
- Time-Trend Analysis
  - Used to see how the firm's performance is changing through time
- Peer Group Analysis
  - Compare to similar companies or within industries

Potential Problems in Financial Analysis

- There is no underlying theory, so there is no way to know which ratios are most relevant.
- Benchmarking is difficult for diversified firms.
- Globalization and international competition makes comparison more difficult because of differences in accounting regulations.
- Regulated monopolies
- Extraordinary, or one-time, events



Discounted Cash Flow Valuation



Outline

- The One-Period Case
- The Multiperiod Case



Discounted Cash Flow Valuation

- A dollar today is more valuable than a dollar to be received in the future

- Why?
  - A dollar today is more valuable because:
    - It can be invested to make more dollars
    - It can be immediately consumed
    - There is no doubt about its receipt

An Important Corollary

- If you know your required rate of return and the length of time before cash is harvested, you can calculate some critical metrics:
  - The value today of a payment to be received in the future
    - **This measure is called a "Present Value"**
  - The value in the future of a sum invested today
    - **This measure is called a "Future Value"**
  - Present and Future Values can be calculated over single and multiple periods

1 The One-Period Case

- If you were to invest $10,000 at 5-percent interest for one year, your investment would grow to $10,500.

$500 would be interest ($10,000 × .05)

$10,000 is the principal repayment ($10,000 × 1)

$10,500 is the total due. It can be calculated as:

$$\$10{,}500 = \$10{,}000 \times (1.05)$$

❑The total amount due at the end of the investment is call the *Future Value* (*FV*).

Future Value

- In the one-period case, the formula for FV can be written as:

$$FV = C_0 \times (1 + r)$$

Where C_0 is cash flow today (time zero), and

r is the appropriate interest rate.



Present Value

- Present Value is today's value of a sum to be received in the future given a specific rate of interest and time horizon.
- Suppose you were promised $10,000 due in one year when interest rates are 5-percent. Your investment be worth $9,523.81 in today's dollars.

$$\$9{,}523.81 = \frac{\$10{,}000}{1.05}$$

The amount that a borrower would need to set aside today to be able to meet the promised payment of $10,000 in one year is the *Present Value* (*PV*).

Note that $\$10{,}000 = \$9{,}523.81 \times (1.05)$.

Present Value

- In the one-period case, the formula for PV can be written as:

$$PV = \frac{C_1}{1+r}$$

Where C_1 is cash flow at date 1, and

r is the appropriate interest rate.

Net Present Value

- The Net Present Value (*NPV*) of an investment is the present value of the expected cash flows, less the cost of the investment.
- Suppose an investment that promises to pay $10,000 in one year is offered for sale for $9,500. Your interest rate is 5%.
- *Should you buy?*



Net Present Value

$$NPV = -\$9{,}500 + \frac{\$10{,}000}{1.05}$$

$$NPV = -\$9{,}500 + \$9{,}523.81$$

$$NPV = \$23.81$$

The present value of the cash inflow is greater than the cost. In other words, the Net Present Value is positive, so the investment should be purchased.

Net Present Value

In the one-period case, the formula for *NPV* can be written as:

$$NPV = -Cost + PV$$

If we had *not* undertaken the positive *NPV* project considered on the last slide, and instead invested our $9,500 elsewhere at 5 percent, our *FV* would be less than the $10,000 the investment promised, and we would be worse off in *FV* terms :

$$\$9{,}500 \times (1.05) = \$9{,}975 < \$10{,}000$$

2 The Multiperiod Case

- The previous examples considered only one period of time.
- It is possible to compute PV & FV for multiple periods of time.
- Doing so requires discrimination between simple and compound interest

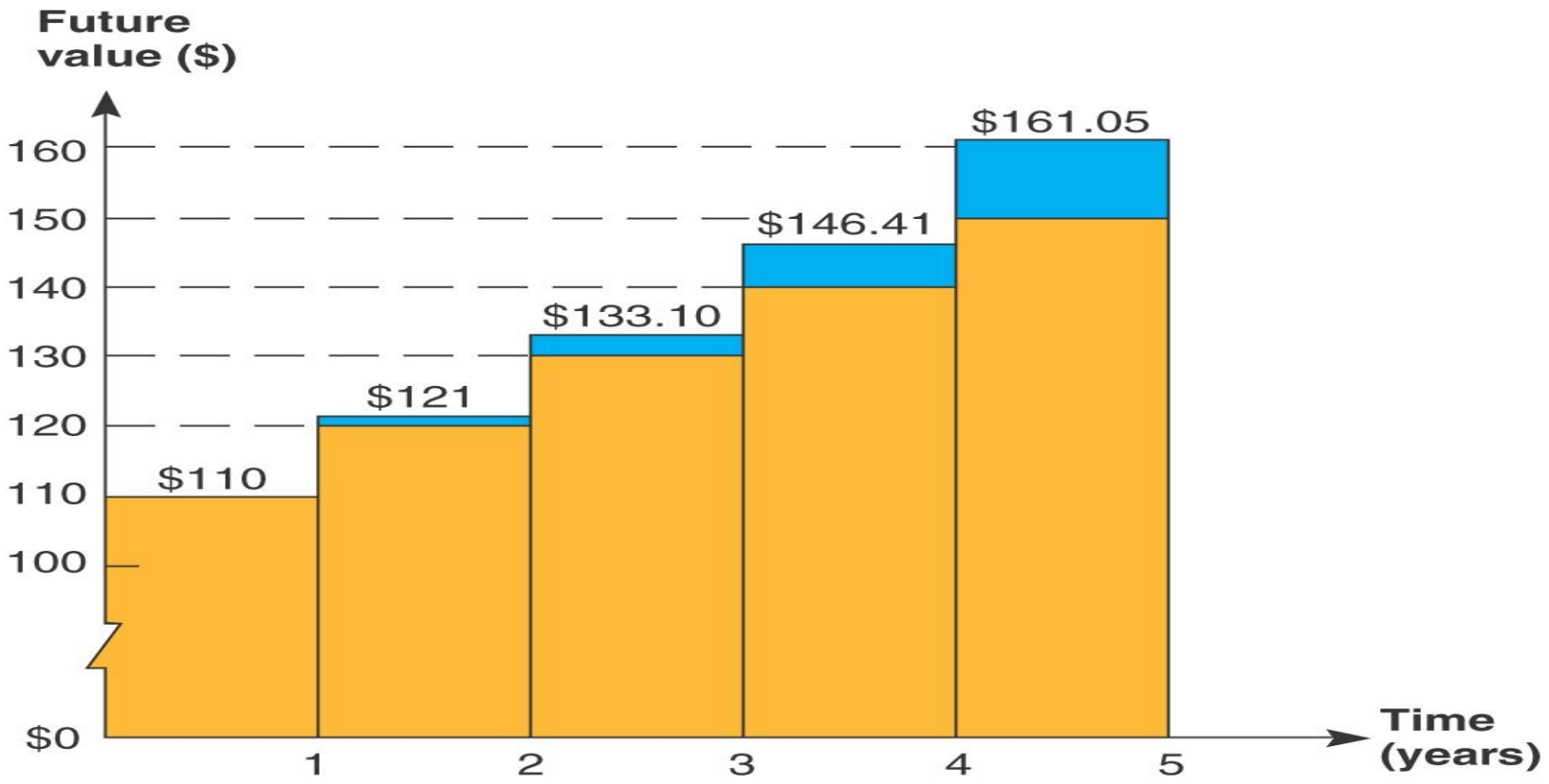


Simple Interest

- Suppose an investor has $1,000 to invest at an interest rate of 9%
- In one year, they will have earned $90 [1000 x 0.09)]
- If they take the $90 earned out of the capital market and reinvest the $1,000 capital at 9% they will again earn $90 in a year's time.
- This process could continue *ad infinitum* with the annual earnings never contributing further to the investor's wealth

Compound Interest

- Suppose an investor has $1,000 to invest at an interest rate of 9%
- In one year, they will have earned $90 (1000 x .09)
- Now suppose that the investor reinvests both the original $1,000 capital AND $90 earnings for another year at 9%
- In one year they will have earned $98.10, an amount $8.10 greater than the previous year.
- In this example, the interest in the second year is higher than the first because it is paid on the initial capital AND prior earnings.
- In other words, the earnings also earn interest or are compounded
- Compounding may not seem very compelling the early years of an investment. But, we will see that it is a very powerful long term force.



Growth of $100 original amount at 10% per year. Blue shaded area represents the portion of the total that results from compounding of interest.

Multiperiod Future Value

- The general formula for the future value of an investment over many periods can be written as:

$$FV = C_0 \times (1 + r)^T$$

Where

C_0 is cash flow at date 0,

r is the appropriate interest rate, and

T is the number of periods over which the cash is invested.



Example: Multiperiod Future Value

- Suppose a stock currently pays a dividend of $1.10, which is expected to grow at 40% per year for the next five years.
- What will the dividend be in five years?

$$FV = C_0 \times (1 + r)^T$$

$$\$5.92 = \$1.10 \times (1.40)^5$$



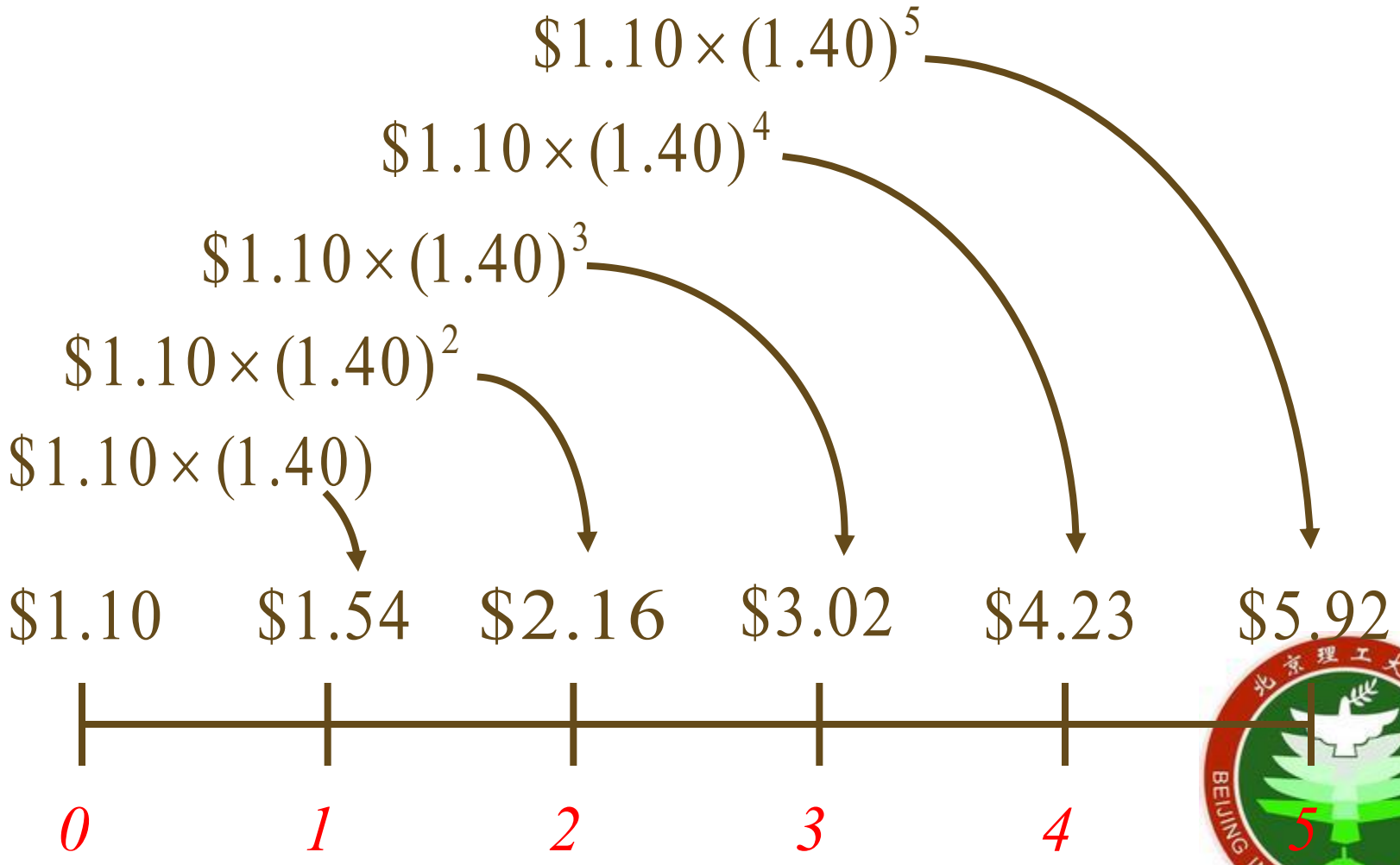
Future Value and Compounding

- Notice that the dividend in year five, $5.92, is considerably higher than the sum of the original dividend plus five increases of 40-percent on the original $1.10 dividend:

$$\$5.92 > \$1.10 + 5 \times [\$1.10 \times .40] = \$3.30$$

This is due to compounding.

Future Value and Compounding



Multiperiod Present Value

- The general formula for the present value of an investment over many periods can be written as:

$$PV = C_T / (1 + r)^T$$

Where

C_T is cash flow at date T,

r is the appropriate interest rate, and

T is the number of periods over which the cash is invested.



Example: Multiperiod Present Value and Discounting

- How much would an investor have to set aside today in order to have $20,000 five years from now if the current rate is 15%?



$$\$9{,}943.53 = \frac{\$20{,}000}{(1.15)^5}$$

Solve for Any Variable

- Examples thus far have offered the time and interest rate and solved for PV or FV
- Keep in mind that there are four variables:
  - PV
  - FV
  - T
  - R
- If you have any three you can solve for the fourth.



Example: Solve for Time

If we deposit $5,000 today in an account paying 10%, how long does it take to grow to $10,000?

$$FV = C_0 \times (1+r)^T \qquad \$10{,}000 = \$5{,}000 \times (1.10)^T$$

$$(1.10)^T = \frac{\$10{,}000}{\$5{,}000} = 2$$

$$\ln(1.10)^T = \ln(2)$$

$$T = \frac{\ln(2)}{\ln(1.10)} = \frac{0.6931}{0.0953} = 7.27 \text{ years}$$



Example: Solve for Required Return

Assume the total cost of a college education will be $50,000 when your child enters college in 12 years. You have $5,000 to invest today. What rate of interest must you earn on your investment to cover the cost of your child's education?

$$FV = C_0 \times (1+r)^T \qquad \$50{,}000 = \$5{,}000 \times (1+r)^{12}$$

$$(1+r)^{12} = \frac{\$50{,}000}{\$5{,}000} = 10 \qquad (1+r) = 10^{1/12}$$

$$r = 10^{1/12} - 1 = 1.2115 - 1 = .2115$$

About 21.15%